MILITARY SPECIFICATION
SETTER, FUZE, M13, M18, M13A1, and M18A1

1. SCOPE

1.1 This specification covers four fuze setters of a single type. (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS
Military

MIL-P-11947	Packaging of Setter, Fuze, M13, W/E and M18, W/E
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of

DRAWINGS
Ordnance Corps

73-3A25-1	Fuze Setter, M13
73-3A42-1	Fuze Setter, M18
F7652345	Fuze Setter, M13A1
F7686755	Fuze Setter, M18A1

(Copies of specifications standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General.- Unless otherwise specified, the requirements common to all fire control instruments as specified in Specification MIL-F-13926 shall be mandatory insofar as they are applicable.

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3.2 Material.- Material shall be in accordance with the applicable drawings and specifications forming a part of this specification.

3.3 Detail

3.3.1 Scales and indexes.- Lines, figures, and indexes shall be clear, distinct, and accurate. Lines and figures requiring translucent finish shall not be opaque when illuminated.

3.3.2 Backlash.- Mating spiral gears, shall have a minimum backlash of 1.36 angular mils.

3.3.3 Adjusting devices.- Adjusting devices shall be locked securely after adjustment has been made.

3.3.4 Holding devices and stops.- Holding devices and stops shall be positive in action.

3.3.5 Drive elements.- Drive elements to the data indicator shall function freely and the backlash shall not exceed the width of a line on the indicator scale.

3.4 Electrical

3.4.1 Voltage input.- The fuze setter shall meet the performance requirements of this specification when 115 volts, 60 cycles is applied. The instrument shall not be damaged when 130 volts, 60 cycles is applied for a period of 2 hours.

3.4.2 Illumination.- The fuze setter dial shall be clearly visible in complete darkness with a minimum of scattered light.

3.4.3 High potential.- There shall be no puncture of insulation, no spark-over, or decrease in dielectric strength of the insulation sufficient to cause failure of the fuze setter when tested in accordance with 4.3.1.

3.4.4 Insulation resistance.- The fuze setter shall have a minimum insulation resistance of 50 megohms for the M13 and M18 and 10 megohms for the M13A1 and M18A1 when tested in accordance with 4.3.2.

3.4.5 Electrical zero.- The electrical index shall be checked for electrical zero in accordance with 4.3.3. When the voltmeter reads approximately 30 volts, the electrical index shall coincide with the mechanical index set at "0" of the scale indicator. When the voltmeter reads approximately 200 volts, the electrical index shall coincide with the mechanical index set at "16" of the scale indicator.

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3.4.6 Continuity.- The fuze setter electrical system shall be tested for continuity and shall conform to the wiring diagram.

3.5 Performance

3.5.1 Functioning.- The action of the fuze setter shall be smooth when, with a master fuze properly inserted, the mechanism completes one complete cycle of operation.

3.5.2 Electrical index.- The electrical index shall not have a variation greater than the width of a line on the indicator scale when the mechanical index is turned one revolution by means of the handwheel.

3.5.3 Accuracy

3.5.3.1 Mechanical.- The readings on the "set" fuze and the readings indicated by the mechanical index shall be the same within a tolerance of plus or minus 0.1 fuze second, with an overall average of less than 0.05 fuze second when tested in accordance with 4.4.1.1.

3.5.3.2 Electrical.- With the mechanical index set at zero and the synchro electrically zeroed, the electrical index shall coincide with the mechanical index within the width of a line on the indicator scale.

3.5.3.3 Repeater.- The error between the fuze value transmitted to the repeater and the value of the electrical index shall be not greater than plus or minus 0.05 fuze second with an overall average of less than 0.03 fuze second when tested in accordance with 4.4.1.2.

3.5.3.4 Setting.- The error between the fuze value transmitted to the repeater, the value of the electrical index, and the value of the set fuze, shall be not greater than plus or minus 0.10 fuze second, with an overall average of less than 0.05 fuze second when tested in accordance with 4.4.1.3.

3.5.4 Torque control.- The fuze setter shall be capable of setting fuzes requiring a torque of 150 pound-inches for the M13 and M13A1, and 250 pound-inches for the M18 and M18A1.

3.6 Environmental

3.6.1 Temperature.- The fuze setter shall meet the performance requirements of this specification throughout the air temperature range extending from plus 51.5°C (+125°F) (minimum exposure of 4 hours with full impact of solar radiation, 360 BTU/Ft. Sq/Hr.) to minus 40°C

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(+40°F) (minimum exposure of 3 days without benefit of solar radiation). The fuze setter shall be capable of safe storage throughout the air temperature range extending from plus 71°C (+160°F) (for periods as long as 4 hours per day) to minus 62.2°C (-80°F) (for periods of at least 3 days duration). Test shall be in accordance with 4.5.2.

3.6.2 Humidity.- The fuze setter shall not suffer damage or fail to meet the performance requirements of this specification immediately after being subjected to the humidity test specified in 4.5.3.

3.6.3 Rain.- The fuze setter shall show no evidence of water penetration, shall not suffer damage, or fail to meet the performance requirements of this specification immediately after being subjected to the rain test specified in 4.5.4.

3.6.4 Vibration.- The fuze setter shall not suffer damage or fail to meet the performance requirements of this specification immediately after being subjected to the vibration test specified in 4.5.5.

3.6.5 Shock.- The fuze setter shall not suffer damage or fail to meet the performance requirements of this specification immediately after being subjected to the shock test specified in 4.5.6.

3.7 Workmanship.- Workmanship shall be in accordance with Specification MIL-F-13926.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Unless otherwise specified, fuze setters shall be inspected for conformance with this specification and with Specification MIL-F-13926. Unless otherwise specified, standard ambient tests shall be performed at a temperature of 60°F to 90°F, a barometric pressure of 28 to 32 inches of mercury, and a relative humidity not greater than 80 percent.

4.2 Detail

4.2.1 Backlash.- Backlash of mating spiral gears shall be measured on the face of the dial after assembly by means of an indicator. The unit shall meet the requirements of 3.3.2.

4.3 Electrical

4.3.1 High potential.- The insulation between ungrounded circuits and ground, and between isolated circuits shall be subjected to a potential of 1000 volts (rms) for the M13 and M18 and 900 volts (rms) for the M13A1 and M18A1, at a frequency of 60 cycles per second. Three of the first five and one out of every additional 50

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or fraction thereof shall be tested. The test voltage shall be applied at a rate of 500 volts (rms) per second until the test potential is reached. The test potential shall be maintained for one minute. For all other units the test voltage shall be applied instantaneously for a period not less than 5 seconds or more than 30 seconds. The unit shall meet the requirements of 3.4.3. Insulation resistance test shall be performed immediately after this test.

4.3.2 Insulation resistance.- The insulation resistance between ungrounded circuits and ground, and between isolated circuits shall be measured by an approved method using a source of d-c potential of 500 volts. The unit shall meet the requirements of 3.4.4.

4.3.3 Electrical zero.- Connections shall be made as shown in figure 1. The unit shall meet the requirements of 3.4.5.

4.4 Performance

4.4.1 Accuracy

4.4.1.1 Mechanical.- A mechanical index value shall be set into the fuze setter. The shell shall be inserted and the fuze shall be set. The reading on the "set" fuze and the mechanical index shall be as specified in 3.5.3.1. A minimum of 10 fuze settings at different positions covering the full scale, selected by the inspector, shall be used.

4.4.1.2 Repeater.- Fuze values shall be introduced into the transmitter increasing from 2 to 30 fuze seconds, and decreasing from 30 to 2 fuze seconds. The unit shall be inspected for a minimum of 10 different settings in each direction selected by the inspector. The electrical index shall repeat the fuze values transmitted as specified in 3.5.3.3.

4.4.1.3 Setting.- Above settings shall be repeated and a fuze shall be set for each setting. The fuze value transmitted to the repeater the value of the electrical index, and the value of the fuze shall be as specified in 3.5.3.4.

4.4.2 Rotation.- Rotation of the electrical index shall be checke by connecting the repeater with a standard transmitter to see that when the transmitter is rotated, the electrical index attached to the rotor shaft of the repeater rotates in the proper direction and without any drag.

4.4.2.1 Terminal sequence

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4.4.2.1.1 The positive direction of rotation of a synchro shaft with respect to the case shall be the direction of phase rotation, which is defined in the specification under which the synchro is procured. Synchros used in fuze setters M13 and M18 have clockwise phase rotation; so, when the synchro is viewed from the shaft extention end, the positive direction of rotation of the shaft is clockwise. Synchros used in fuze setters M13A1 and M18A1 have counterclockwise phase rotation; so the positive direction of rotation of the shaft is counterclockwise, as seen from the shaft extension end.

4.4.2.1.2 Pin No. 4 of fuze setter receptacle and transmitter terminal R1 (sometimes designated 4 or X) shall be connected to one side of the power source; pin No. 5 of the fuze setter receptacle and transmitter terminal R2 (sometimes designated 5 or Y) shall be connected to the other side of the power source. If the data represented by the shaft of the transmitter position increases as the shaft is rotated in the positive direction, the stator terminals S1, S2 and S3 (sometimes designated 1, 2 and 3) shall be connected to the receptacle of the fuze setter in normal order. If the data decreases as the shaft is rotated positively, terminals S1, S2 and S3 shall be connected to the receptacle of the fuze setter in reverse order.

4.5 Environmental

4.5.1 General.- The environmental tests specified shall be limited to those contained in the following sub-paragraphs. Three of the first five fuze setters and one out of every additional 100 or fraction thereof shall be tested.

4.5.2 Temperature.- The fuze setter shall be exposed to an air temperature of minus 62.2°C (-80°F) for a period of 3 days. The air temperature shall then be raised to minus 40°C (-40°F) and after the internal temperature of the unit has reached equilibrium, it shall be tested for conformance with 3.6.1. The air temperature shall be raised to plus 71°C (+160°F) and after the internal temperature of the unit has reached equilibrium at that temperature it shall be allowed to remain at that temperature for a period of 4 hours. The air temperature shall be reduced to plus 51.5°C (+125°F) and after the internal temperature of the unit has reached equilibrium at that temperature the top surface shall be exposed to 360 BTU/Ft. Sq/Hr., for a period of 4 hours. The unit shall then be tested for conformance with 3.6.1.

4.5.3 Humidity.- A test chamber that is capable of beind sealed shall be used; the source of heat shall be arranged in such a manner that the radiant heat will not fall upon the fuze setter tested. A

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unit shall be placed in the chamber, the temperature of which is adjusted to plus 46.1°C ± 1.1°C (+115°F ± 2°F) and the relative humidity adjusted to 95 ± 5 percent. The unit shall remain under the above conditions for a period of 6 hours. The temperature of the chamber shall be readjusted to plus 26.7°C ± 1.1°C (+80°F ± 2°F) and the unit shall be allowed to remain under this new condition for a period of 6 hours. The transitional period shall be 2 hours; the rate of change during that period shall be uniform. The above cycle shall be repeated 10 times. Immediately after this test the unit shall be tested for conformance with 3.6.2.

4.5.4 Rain.- The fuze setter shall be mounted in the test chamber to simulate installation conditions. The rain test chamber temperature shall be maintained between 20°C and 30°C (68°F and 86°F) throughout the test period. A simulated rainfall of 4 + 1 inch per hour shall be produced by a suitable means of such design that the water is emitted in the form of small droplets rather than a fine mist. The temperature of the water shall be maintained between 11°C and 24°C (51.8°F and 75.2°F). The water spray shall be directed on the fuze setter for a period of 2 hours in directions ranging from vertical to 45° from the vertical. The simulated rain shall be directed on all surfaces of the fuze setter that are normally subjected to rain when the instrument is in its operating position. Immediately after this test the unit shall be tested for conformance with 3.6.3.

4.5.5 Vibration.- The fuze setter shall be mounted rigidly using its own mounting provisions to simulate field use and shall be vibratted successively along each of the three mutually perpendicular directions. Each unit tested shall be subjected to a simple harmonic motion having an amplitude of 0.031 inch (0.062 inch total excursion) at 55 cycles per second. The entire range of frequencies from 10 to 55 cycles per second and return to 10 cycles per second shall be traversed in one minute. Immediately after this test the unit shall be tested for conformance with 3.6.4.

4.5.6 Shock.- The fuze setter shall be mounted rigidly using its own mounting provisions to simulate field use. A shock of 20 g's shall be imparted three times in each direction along each of the three mutually perpendicular directions making a total of 18 shock impulses. Each shock impulse shall have a time duration of 11±1 milliseconds; the maximum g value shall occur at approximately 5.5 milliseconds and shall be within ± 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental from the measured shock impulse. Immediately after this test the unit shall be tested for conformance with 3.6.5.

5. PREPARATION FOR DELIVERY

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5.1 Packaging, packing, and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-11947.

6. NOTES

6.1 Intended use.- The fuze setters M13 and M13A1 are used as part of the fire control system on the 90MM Antiaircraft Gun Mounts and the fuze setters M18 and M18A1 are used as part of the fire control system on the 120MM Antiaircraft Gun Mounts. The fuze setters are mechanically operated devices for setting time fuzes in accordance with electrical data received from the director.

6.2 Ordering data.- Contracts and invitations for bids should specify the title, number, date of this specification, and type of packing.

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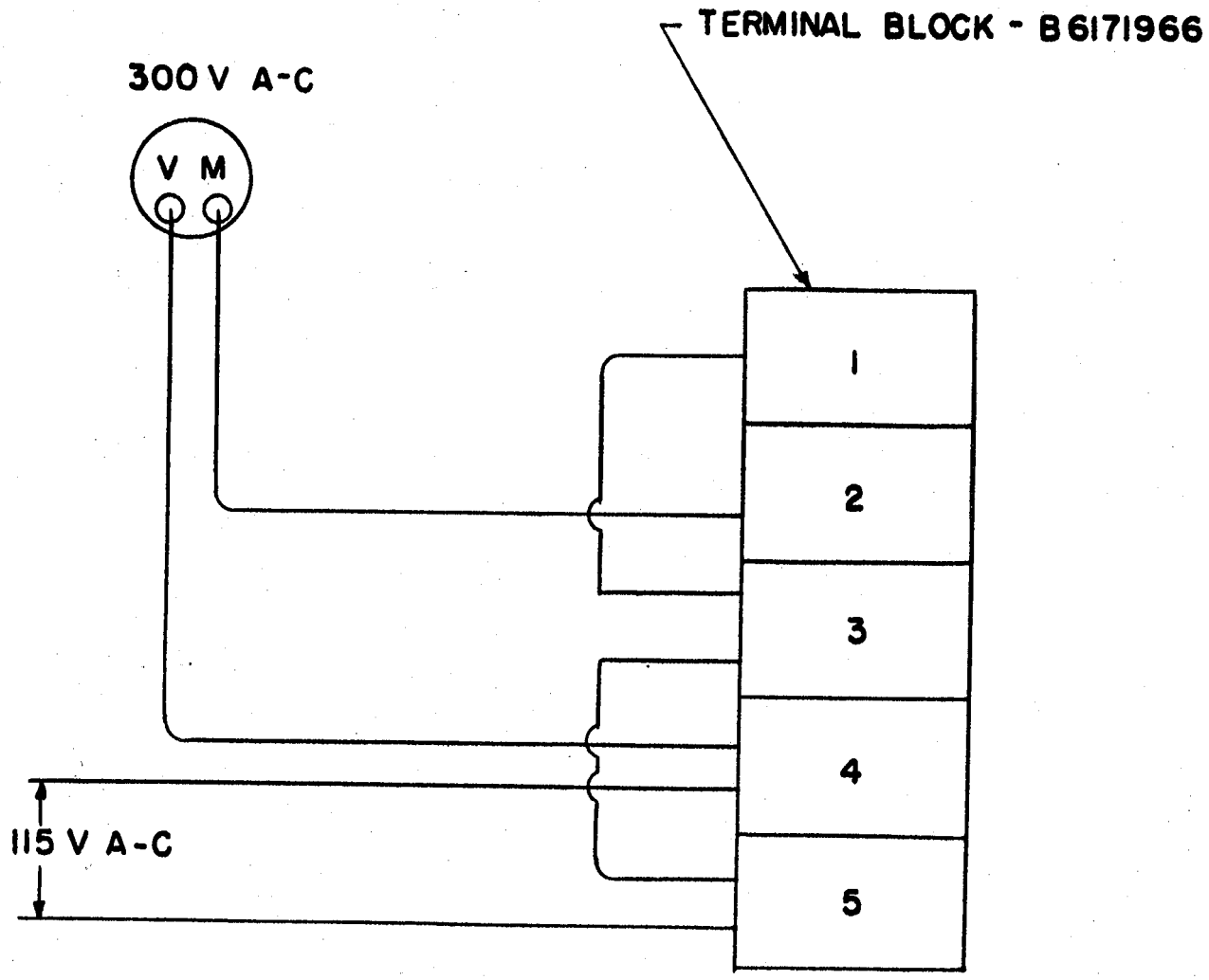


FIGURE 1

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